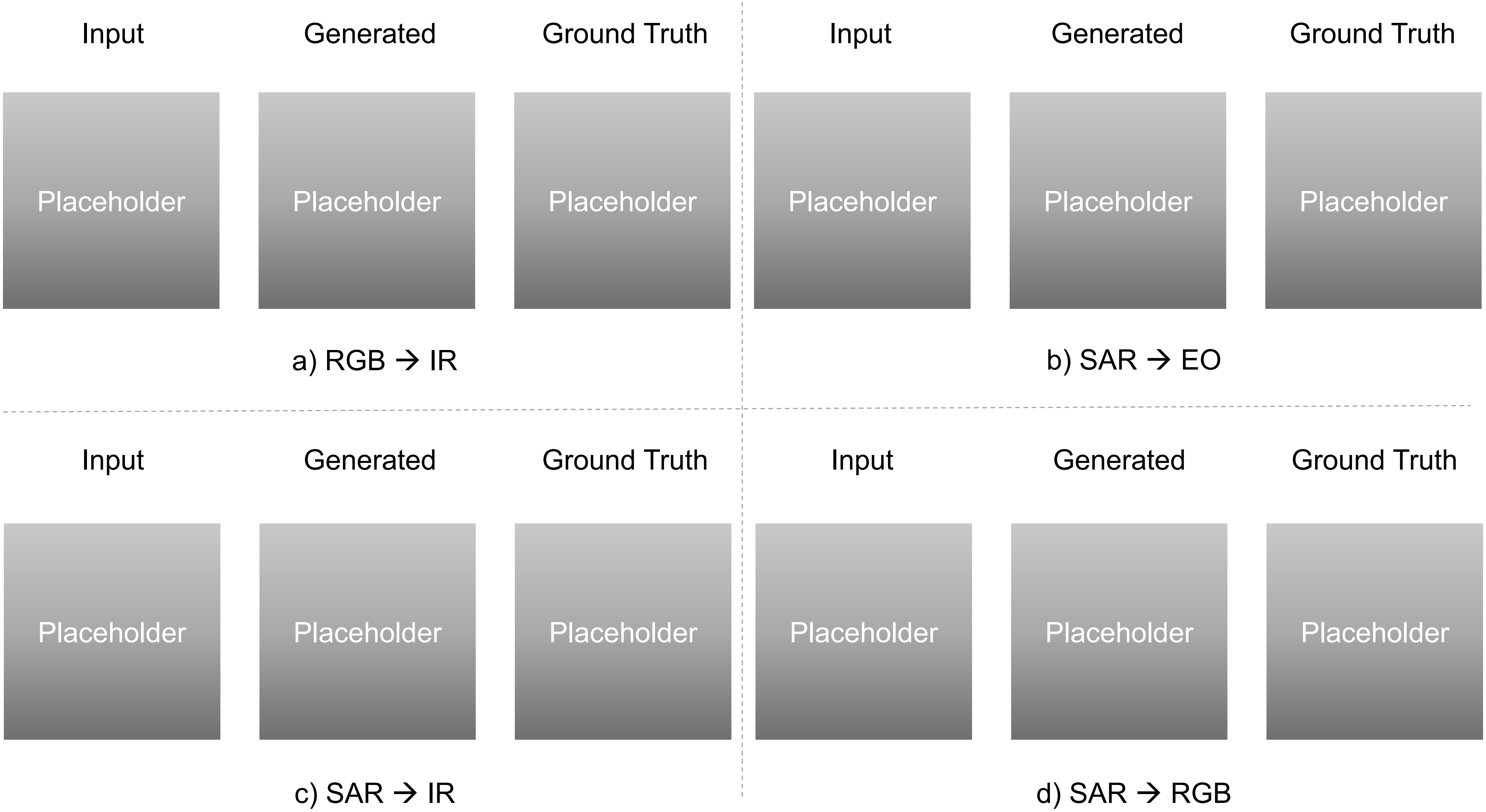
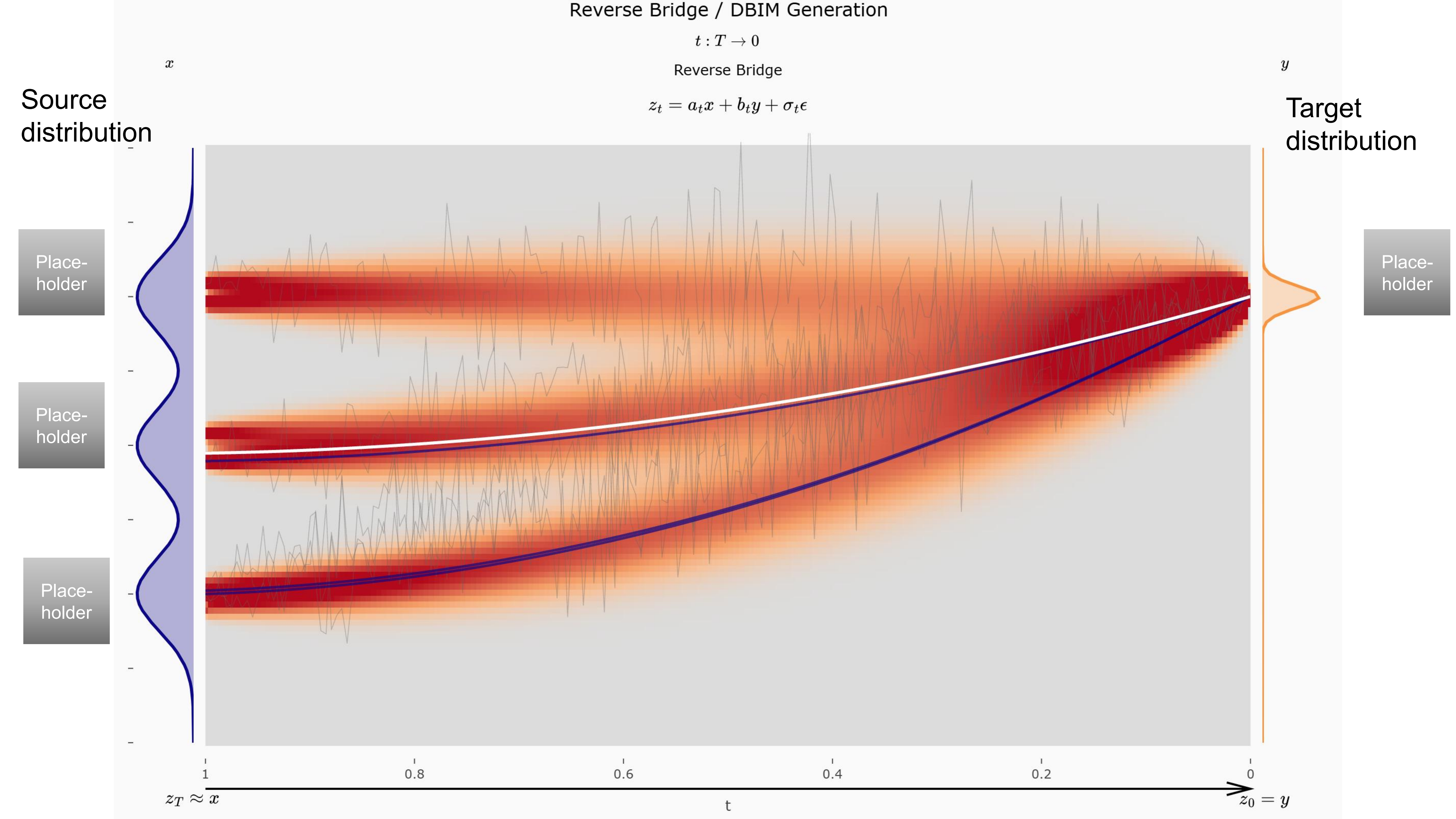
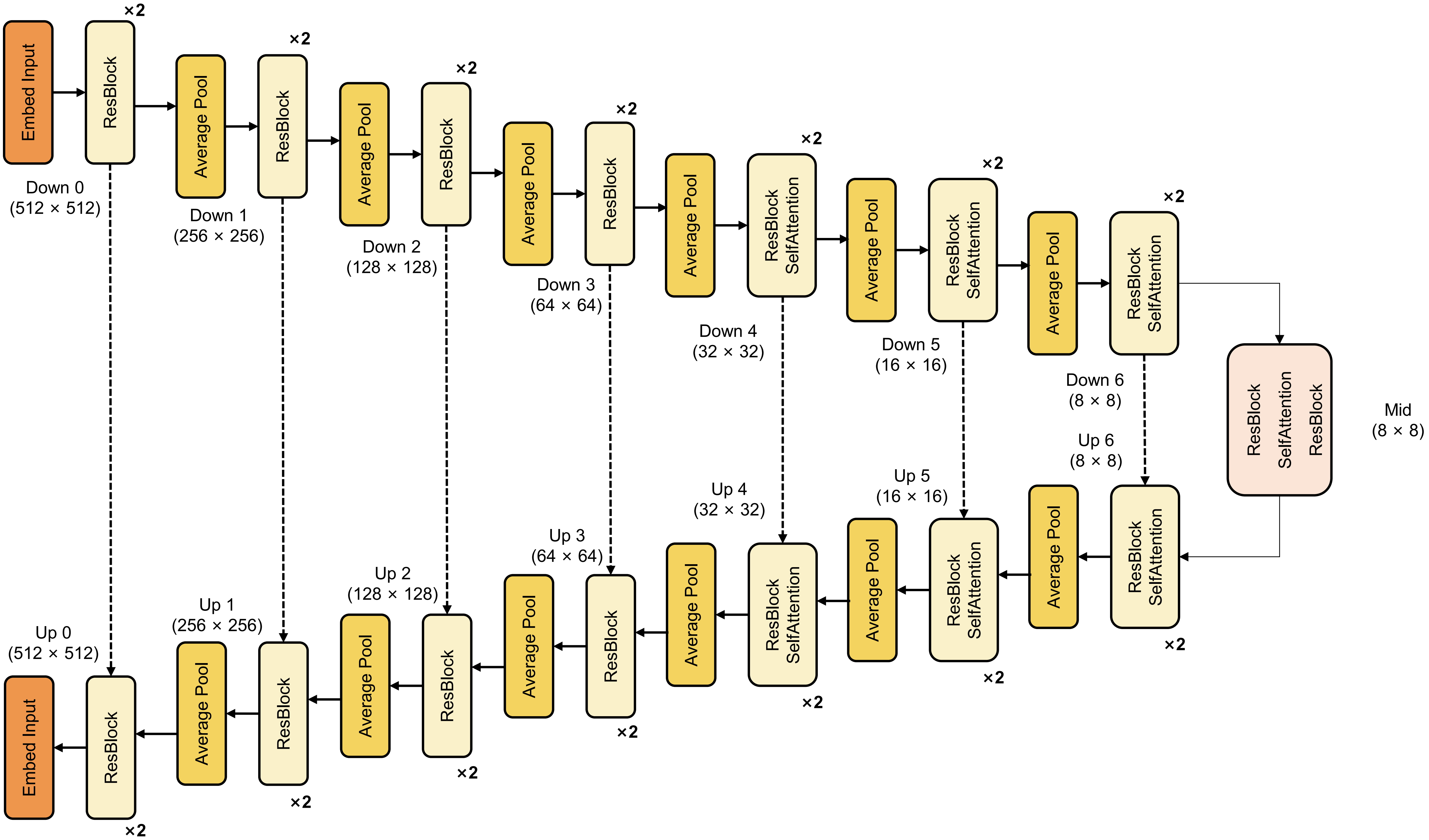
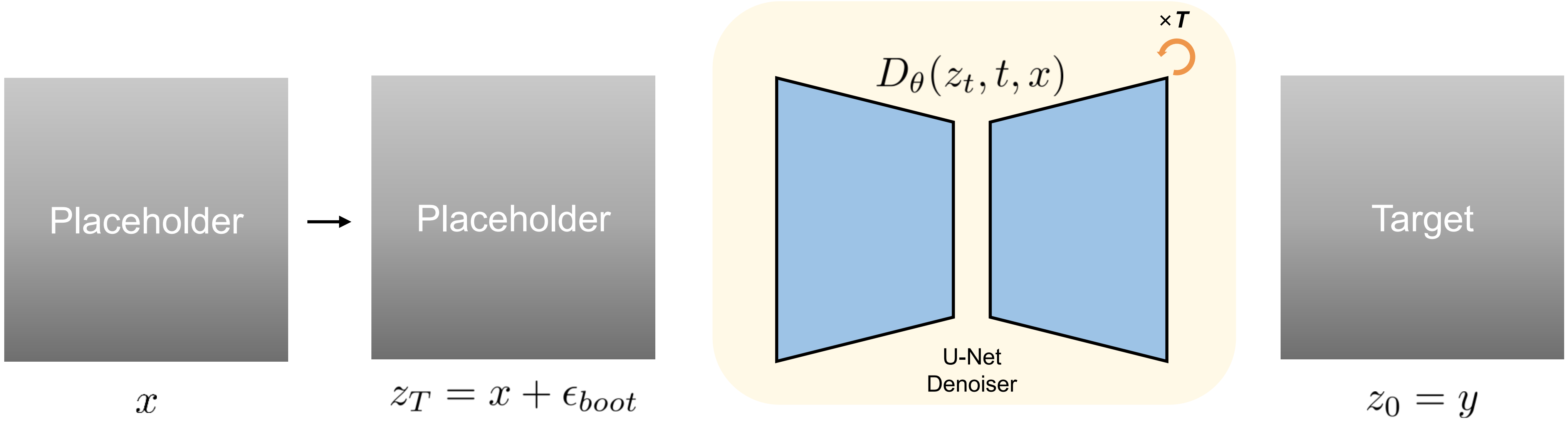
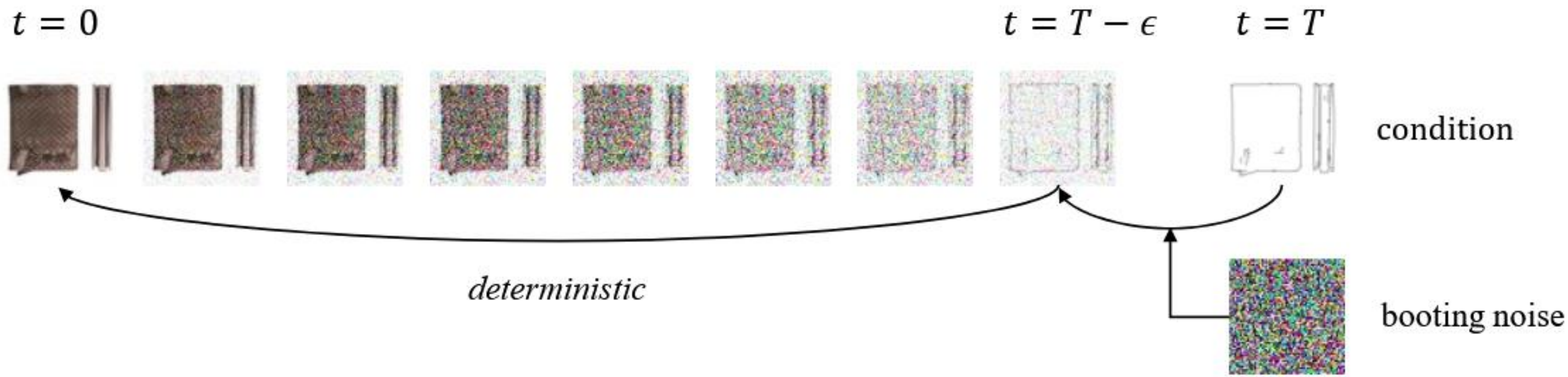

Input
Generated
Ground Truth
Placeholder
Placeholder
Placeholder
a) RGB → IR
Input
Generated
Ground Truth
Placeholder
Placeholder
Placeholder
b) SAR → EO
Input
Generated
Ground Truth
Placeholder
Placeholder
Placeholder
c) SAR → IR
Input
Generated
Ground Truth
Placeholder
Placeholder
Placeholder
d) SAR → RGB



Reverse Bridge / DBIM Generation
$t : T \to 0$
x
Reverse Bridge
y
$z_t = a_t x + b_t y + \sigma_t \epsilon$
Source distribution
Target distribution
Place-holder
Place-holder
Place-holder
Place-holder
1
0.8
0.6
0.4
0.2
0
$z_T \approx x$
t
$z_0 = y$


Embed Input
ResBlock
×2
Down 0
(512 × 512)
Average Pool
ResBlock
×2
Down 1
(256 × 256)
Average Pool
ResBlock
×2
Down 2
(128 × 128)
Average Pool
ResBlock
×2
Down 3
(64 × 64)
Average Pool
ResBlock
SelfAttention
×2
Down 4
(32 × 32)
Average Pool
ResBlock
SelfAttention
×2
Down 5
(16 × 16)
Average Pool
ResBlock
SelfAttention
×2
Down 6
(8 × 8)
ResBlock
SelfAttention
ResBlock
Mid
(8 × 8)
Up 6
(8 × 8)
ResBlock
SelfAttention
×2
Average Pool
Up 5
(16 × 16)
ResBlock
SelfAttention
×2
Average Pool
Up 4
(32 × 32)
ResBlock
SelfAttention
×2
Average Pool
Up 3
(64 × 64)
ResBlock
×2
Average Pool
Up 2
(128 × 128)
ResBlock
×2
Average Pool
Up 1
(256 × 256)
ResBlock
×2
Average Pool
Up 0
(512 × 512)
ResBlock
×2
Embed Input


Placeholder
Placeholder
$D_\theta(z_t, t, x)$
$\times T$
U-Net
Denoiser
Target
x
$z_T = x + \epsilon_{boot}$
$z_0 = y$


$t = 0$
$t = T - \epsilon$
$t = T$
condition
deterministic
booting noise

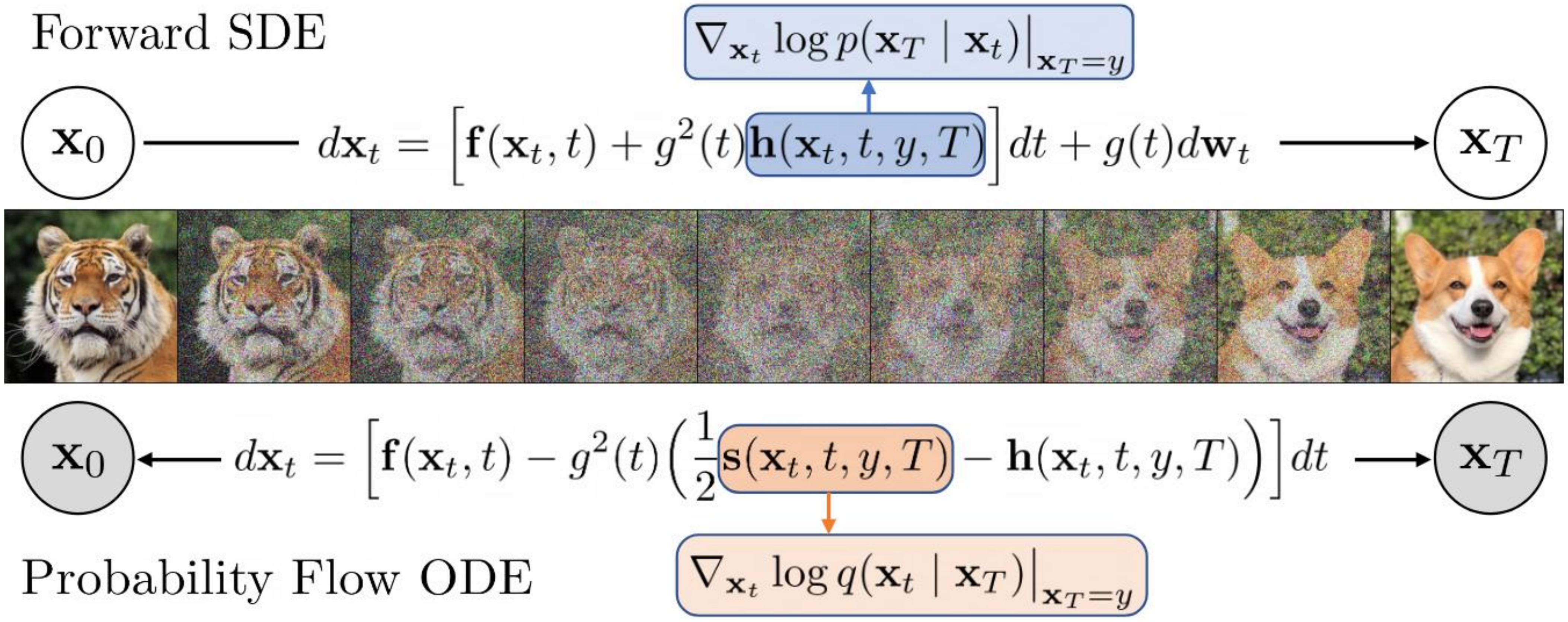


Figure 1: **A schematic for Denoising Diffusion Bridge Models.** DDBM uses a diffusion process guided by a drift adjustment (in blue) towards an endpoint $\mathbf{x}_T = y$. They lears to reverse such a bridge process by matching the denoising bridge score (in orange), which allows one to reverse from $\mathbf{x}_T$ to $\mathbf{x}_0$ for any $\mathbf{x}_T = \mathbf{y} \sim q_{\text{data}}(\mathbf{y})$. The forward SDE process shown on the top is unidirectional while the probability flow ODE shown at the bottom is deterministic and bidirectional. White nodes are stochastic while grey nodes are deterministic.